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Sahaj Saliya

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https://sahajivvix-1.github.io/Portfolio2026/

Education

Pandit Deendayal Energy University - PDEU (Formerly PDPU) — Gandhinagar, GJ
B.Tech in Information & Communication Technology (ICT), ***GPA: 8.5/10.0*** — *June 2023 – May 2027*

Projects

OpenEnv — RL Data Cleaning Agent — GitHub
Reinforcement Learning · Large Language Models (LLMs) — *2026*

- Developed an autonomous data cleaning agent using **Llama 3** and Reinforcement Learning to intelligently transform messy tabular data based on reward signals.
- Built a custom RL environment (OpenEnv) featuring reward shaping, anti-loop penalties, and automated state observations for context-aware cleaning actions.
- Containerized the full stack with **multi-stage Docker** and deployed on HuggingFace Spaces with a FastAPI-based REST API for inference.
- **Tech Stack:** Python, Reinforcement Learning, Llama 3 (OpenRouter), Docker, FastAPI, HuggingFace Spaces.

DCGAN Modernization & Analytics Pipeline — GitHub
Generative AI & Post-Training Diagnostics — *2026*

- Implemented a high-stability Generative Adversarial Network (GAN) using 2018-2022 training paradigms including Label Smoothing and D-Dropout to stabilize the Nash Equilibrium between the Generator and Discriminator.
- Developed a professional post-training analytics dashboard that automates evaluation through confidence histograms ($D(x)$ vs $D(G(z))$) and real-time visualization of learned edge/texture detectors.
- **Tech Stack:** PyTorch, Torchvision, Python 3, Advanced Convolutions, Latent Space Analytics, Data Visualization.

SentinelProxy Suite — GitHub
Secure Network Gateways · Traffic Intelligence — *2025*

- Developed an AI-enhanced HTTPS proxy and DNS resolver featuring real-time traffic inspection, role-based access control (RBAC), and dynamic blocklist enforcement.
- Automated Windows network orchestration via PowerShell and integrated a modular Node.js backend with SQLite for persistent auditing and logging.
- **Tech Stack:** Node.js, Socket.IO, SQLite, http-proxy, node-forge, PowerShell, JavaScript.

Microplastic Detection
Visual Intelligence · Automated Environmental Analysis — *2025*

- Trained and fine-tuned a **YOLOv11** object detection model to detect microplastic particles from visual data, optimizing it to classify microplastics while estimating particle size under varying imaging conditions.
- **Tech Stack:** Python, LLM, YOLOv11, PyTorch, OpenCV, CUDA, NumPy, Computer Vision.

Specialized Python Applications — GitHub
Production Automation · Media Processing · System Utilities — *2024 – 2025*

- **vMix Manager:** A direct XML production engine for vMix that allows batch processing of inputs, category migrations, and asset relinking outside the native GUI.
- **PyEnv Launcher:** A sleek PyQt6 graphical interface for managing PyEnv environments, virtual environments, and dependency isolation for complex Python projects.
- **Video & PDF Suite:** A collection of media tools including an FFmpeg-powered folder analyzer and an OpenCV-based PDF-to-video scrolling generator.
- **Smaran Reminder:** A non-intrusive desktop automation tool with deep scheduling logic and customizable popup bursts built with PyQt6.
- **Tech Stack:** Python, PyQt6, FFmpeg, OpenCV, XML Parsing, JSON, Docker.

Skills

Programming & Core Tools: C++ (OOP's), Python (Libraries), MATLAB, NumPy, Pandas, Scikit-Learn
Machine Learning & AI: Deep Learning, Reinforcement Learning (RL), LLM's, EDA, Probability & Statistics, Linear Algebra, PyTorch, Data Visualization (Power BI, Excel (VBA), Matplotlib, Seaborn)
System Design & Databases: Scalable System Design, Microservices, NoSQL, MongoDB, MySQL, PostgreSQL, SQLite
Computer Networks: Git/GitHub, Virtualization, OS Managment, SysAdmin, Docker, Linux

Certifications

- **Deep Learning**: NPTEL – IIT Ropar, **Score: 66%**
- **AI/ML for Geodata Analysis**: IISRO-IIRS (ISRO) – Applied AI/ML to satellite imagery and geospatial data.
- **Code4Cause Hackathon 2.0**: National-level hackathon participation, solving real-world technology problems under time constraints.
- **Signal Processing Onramp**: Applied MATLAB to real-world signal processing, visualization, and data analysis.